The Adjoint Based Subring and Universal Grading Theorem 3.6.4

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Private reading transcription of the 2015 author final copy
with a separately identified 2026 author clarification

Source and selected scope. *Tensor Categories*, AMS Mathematical Surveys and Monographs 205 (2015), author final copy, 362 physical pages. The sole primary selection is Theorem 3.6.4 on physical page 77, printed page 60, with its complete printed proof on physical page 78, printed page 61. Propositions 3.6.2, Lemma 3.6.3, and Corollary 3.6.6 are complete uncounted support and consequences of the same construction. The original opening unital context, the later wording "based ring," and the finite-rank aside are retained in their original positions. No global finite-rank condition is inserted.
The source is `https://math.mit.edu/~etingof/egnobookfinal.pdf`. The separate author corrections dated February 14, 2026 are retained at `https://math.mit.edu/~etingof/tensbookcor.pdf`. The selected clarification of the based-module dependency is displayed separately below. It is not silently substituted into the 2015 text.

Selected definitions and coefficient identity

Source: Physical page 66, printed page 49; Definitions 3.1.1 and 3.1.3, Remark 3.1.2 and equation (3.1).

Let $\mathbb{Z}_+$ denote the semi-ring of non-negative integers.

Definition 3.1.1. Let A be a ring which is free as a $\mathbb{Z}$-module.

(i) A $\mathbb{Z}_+$-basis of A is a basis $B = \{b_i\}_{i\in I}$ such that $b_i b_j = \sum_{k\in I} c_{ij}^k b_k$, where $c_{ij}^k \in \mathbb{Z}_+$.

(ii) A $\mathbb{Z}_+$-ring is a ring with a fixed $\mathbb{Z}_+$-basis and with identity 1 which is a non-negative linear combination of the basis elements.

(iii) A unital $\mathbb{Z}_+$-ring is a $\mathbb{Z}_+$-ring such that 1 is a basis element.

Remark 3.1.2. Every $\mathbb{Z}_+$-ring is assumed to have an identity, but according to our terminology a $\mathbb{Z}_+$-ring is not necessarily unital.

Let A be a $\mathbb{Z}_+$-ring, and let I_0 be the set of $i \in I$ such that b_i occurs in the decomposition of 1. Let $\tau : A \to \mathbb{Z}$ denote the group homomorphism defined by

$$\tau(b_i) = \begin{cases} 1 & \text{if } i \in I_0, \\ 0 & \text{if } i \notin I_0. \end{cases} \tag{3.1}$$

Definition 3.1.3. A $\mathbb{Z}_+$-ring A with basis $\{b_i\}_{i\in I}$ is called a based ring if there exists an involution $i \mapsto i^*$ of I such that the induced map

$$a = \sum_{i\in I} a_i b_i \longmapsto a^* = \sum_{i\in I} a_i b_{i^*}, \qquad a_i \in \mathbb{Z},$$

is an anti-involution of the ring A, and

$$\tau(b_i b_j) = \begin{cases} 1 & \text{if } i = j^*, \\ 0 & \text{if } i \neq j^*. \end{cases} \tag{3.2}$$

Source: Physical pages 66–67, printed pages 49–50; complete Proposition 3.1.6 and its printed proof, followed by Definition 3.1.7.

Proposition 3.1.6. *In any based ring, the number $c_{ij}^{k^*}$ is invariant under cyclic permutations of i, j, k.*

Proof. It is easy to see that $c_{ij}^{k^*} = \tau(b_i b_j b_k)$, and $\tau(xy) = \tau(yx)$. Thus, $c_{ij}^{k^*}$ is cyclically symmetric. □

Definition 3.1.7. A multifusion ring is a based ring of finite rank. A fusion ring is a unital based ring of finite rank.

Source: Physical page 73, printed page 56; complete Definitions 3.4.1 and 3.4.2.

Definition 3.4.1. Let A be a $\mathbb{Z}_+$-ring with basis $\{b_i\}$. A $\mathbb{Z}_+$-module over A is an A-module M with a fixed $\mathbb{Z}$-basis $\{m_l\}_{l \in L}$ such that all the structure constants a_{il}^k (defined by the equality $b_i m_l = \sum_k a_{il}^k m_k$) are non-negative integers.

The direct sum of $\mathbb{Z}_+$-modules is also a $\mathbb{Z}_+$-module whose basis is the union of the bases of the summands. We say that a $\mathbb{Z}_+$-module is indecomposable if it is not isomorphic to a nontrivial direct sum of $\mathbb{Z}_+$-modules.

Definition 3.4.2. A $\mathbb{Z}_+$-module M over a $\mathbb{Z}_+$-ring A is called irreducible if it has no proper $\mathbb{Z}_+$-submodules (in other words, the $\mathbb{Z}$-span of any proper subset of the basis of M is not an A-submodule).

Source: Physical page 74, printed page 57; original dependency, Exercise 3.4.3(i).

Exercise 3.4.3(i), as printed in 2015. Prove that a $\mathbb{Z}_+$-module over a based ring is irreducible if and only if it is indecomposable.

Source: Separate corrections dated February 14, 2026, physical page 2; selected Section 3.4 clarification and first correction to Exercise 3.4.3(i).

> *Section 3.4.* The definition of a based $\mathbb{Z}_+$-module over a based ring is accidentally omitted, although it is used further in the text (first time in Proposition 3.6.2).
>
> By definition, a based $\mathbb{Z}_+$-module over a based ring A is one where if $x \in A$ acts by a matrix X then x^* acts by X^T.
>
> *Exercise 3.4.3(i).* The $\mathbb{Z}_+$-module should assumed be based.

Editorial note. This is the selected based-module clarification relevant to the later based-module text. The full separately retained corrections also contain the authors' non-based examples and an alternative finite-rank condition with Lemma 2.1 and its proof on page 3. That alternative is not imposed as a global finite-rank hypothesis here.

Source: Physical page 75, printed page 58; complete Definition 3.5.1 and the following grading/faithfulness paragraphs, before the fusion-ring dimension specialization.

Let A be a unital based ring with a $\mathbb{Z}_+$-basis $B = \{b_i\}_{i \in I}$.

Definition 3.5.1. Let G be a group. A grading of A by G is a partition $B = \bigsqcup_{g\in G} B_g$ of the basis of A into disjoint subsets such that for all $g, h \in G$ and for all basis elements $b_i \in B_g$ and $b_j \in B_h$ the product $b_i b_j$ is a $\mathbb{Z}_+$-linear combination of elements of B_{gh}.

Note that if $b_i \in B_g$ then $b_i^* \in B_{g^{-1}}$. We will call B_g the component of B corresponding to $g \in G$, and B_1 (where 1 denotes the identity element of G) the trivial component of B. Note that the trivial component includes all basis elements contained in the decomposition of $b_i b_i^*$, $i \in I$. In particular, the trivial component contains b_i for all $i \in I_0$.

If A is G-graded, we have a decomposition

$$A = \bigoplus_{g\in G} A_g,$$

where A_g is the $\mathbb{Z}$-span of B_g, $g \in G$. We will say that A_g is the component of A corresponding to g. The subring A_1 will be called the trivial component of A.

The grading is called faithful if all the components B_g, $g \in G$, are nonempty.

3.6. The adjoint based subring and universal grading

Source: Physical page 77, printed page 60; original section opening and all support before Theorem 3.6.4.

Let A be a unital based ring with a $\mathbb{Z}_+$-basis $B = \{b_i\}_{i\in I}$.

Definition 3.6.1. The adjoint subring $A_{\mathrm{ad}} \subset A$ is the minimal based subring of A with the property that $b_i b_i^*$ belongs to A_{ad} for all $i \in I$; i.e., A_{ad} is generated by all basis elements of A contained in $b_i b_i^*$, $i \in I$.

If A is of finite rank (i.e., a fusion ring), let us define $J := \sum_{i\in I} b_i b_i^*$ (note that this is a central element of A by Proposition 3.1.8 for $x = 1$). Then A_{ad} can be alternatively defined as the $\mathbb{Z}$-linear span of basis elements contained in J^n, $n = 0, 1, 2, \ldots$.

Proposition 3.6.2. *Any one-sided (i.e., left or right) based A_{ad}-submodule $M \subset A$ of a based ring A is automatically an A_{ad}-sub-bimodule.*

Proof. It suffices to prove the statement for left modules. We have

$$b_k b_i b_{i^*} = \sum_{r: b_r \in b_k b_i} c_{ki}^r b_r b_{i^*} = \sum_{r: b_r \in b_k b_i} c_{r^* k}^{i^*} b_r b_{i^*}$$

(where $b_r \in b_k b_i$ means that b_r occurs in the decomposition of $b_k b_i$). On the other hand,

$$\sum_{r,i} c_{r^* k}^{i^*} b_r b_{i^*} = \left(\sum_r b_r b_{r^*}\right) b_k.$$

But

$$\sum_{r,i} c_{r^* k}^{i^*} b_r b_{i^*} = \sum_{r: b_r \in b_k b_i} c_{r^* k}^{i^*} b_r b_{i^*} + E_{ki},$$

where E_{ki} is a non-negative linear combination of basis elements. Hence,

$$\left(\sum_r b_r b_{r^*}\right) b_k = b_k b_i b_{i^*} + E_{ki}.$$

So if $b_k \in M$ then $b_k b_i b_{i^*} \in M$ for all i, as desired. □

Let A be a based ring. We can view A as a based $\mathbb{Z}_+$-bimodule over A_{ad}. As such, it decomposes into a direct sum of indecomposable based A_{ad}-bimodules: $A = \bigoplus_{a\in G} A_a$, where G is the index set. This decomposition is unique up to a permutation of G. We may assume that there is an element $1 \in G$ such that $A_1 = A_{\mathrm{ad}}$. Note that $(A_a)^* = \{X^* \mid X \in A_a\}$, $a \in G$, is an indecomposable based A_{ad}-submodule of A and hence $(A_a)^* = A_{a^*}$ for some $a^* \in G$.

Lemma 3.6.3. *For all $x, y \in A_a$, $a \in G$, we have $xy^* \in A_{\mathrm{ad}}$.*

Proof. We may assume that x, y are basis vectors. Observe that

$$M_y = \mathrm{span}\{X \in A_a \mid X \text{ is contained in } b_{i_1}b_{i_1}^* \cdots b_{i_n}b_{i_n}^* y \text{ for some } n, i_1, \ldots, i_n\}$$

is a based left A_{ad}-submodule of A_a. Since A_a is an indecomposable based A_{ad}-bimodule, by Exercise 3.4.3(i), it is irreducible, and so it follows from Proposition 3.6.2 that it is in fact irreducible as a based left A_{ad}-module. Thus, $M_y = A_a$. Therefore, $x \in M_y$, so x is contained in $b_{i_1}b_{i_1}^* \cdots b_{i_n}b_{i_n}^* y$ for some $n, i_1, \ldots, i_n$, i.e., xy^* is a summand in $b_{i_1}b_{i_1}^* \cdots b_{i_n}b_{i_n}^* yy^* \in A_{\mathrm{ad}}$, as required. □

Source: Physical page 77, printed page 60; complete Theorem 3.6.4.

Theorem 3.6.4. *There is a canonical group structure on the index set G with the multiplication defined by the following property:*

$$\begin{gathered} ab = c \quad \text{if and only if } x_a x_b \in A_c, \\ \text{for all } x_a \in A_a,\ x_b \in A_b,\ a, b, c \in G. \end{gathered} \tag{3.6}$$

The identity of G is 1 and the inverse of $a \in G$ is a^*.

Source: Physical page 78, printed page 61; complete printed proof.

Proof. We need to check that the binary operation in (3.6) is well defined. Let $a, b \in G$ and let $x_a, y_a \in A_a$, $x_b, y_b \in A_b$ be basis elements of A.

Suppose that the product $x_a x_b$ contains a basis element $x_c \in A_c$ and the product $y_a y_b$ contains a basis element $y_d \in A_d$ for $c \neq d$. By Lemma 3.6.3, there is a positive integer n and indices $i_1, \ldots, i_n$ such that the element $z := x_a b_{i_1} b_{i_1}^* \cdots b_{i_n} b_{i_n}^* y_a^*$ contains $x_a x_b y_b^* y_a^* = (x_a x_b)(y_a y_b)^*$. But by Proposition 3.6.2, the left based A_{ad}-module generated by x_a is also a right based A_{ad}-module, which means that $x_a b_{i_1} b_{i_1}^* \cdots b_{i_n} b_{i_n}^*$ is a summand in a sum of elements of the form $b_{j_1} b_{j_1}^* \cdots b_{j_m} b_{j_m}^* x_a$, and hence $z \in A_1$. Therefore, $Y := x_c y_d^*$ is in A_1. Multiplying both sides of the last equality by y_d on the right, we conclude that Yy_d is in $A_c \cap A_d$, a contradiction.

Thus, $x_a x_b$ and $y_a y_b$ both belong to the same component A_c and so the binary operation (3.6) is well-defined. It is easy to see that it defines a group structure on G. □

Source: Physical page 78, printed page 61; complete Definition 3.6.5 and Corollary 3.6.6, stopping before the upper-central-series paragraph.

Definition 3.6.5. We will call the grading $A = \bigoplus_{a\in G} A_a$ constructed in Theorem 3.6.4 the universal grading of A. The group G will be called the universal grading group of A and denoted by $U(A)$.

Corollary 3.6.6. *Every based ring A has a canonical faithful grading by the group $U(A)$. Any other faithful grading of A by a group G is determined by a surjective group homomorphism $\pi : U(A) \to G$.*

Proof. Let $A = \bigoplus_{g\in G} A^g$ be a faithful grading of A. Since for every basis element $X \in A$ we have $XX^* \in A^1$, it follows that A^1 contains A_{ad} as a based subring. Hence, each A^g is a based A_{ad}-submodule of A. This means that every component A_a, $a \in U(A)$, of the universal grading $A = \bigoplus_{a\in U(A)} A_a$ of A belongs to some $A^{\pi(a)}$ for some well-defined $\pi(a) \in G$. Clearly, the map $a \mapsto \pi(a)$ is a surjective homomorphism. □

Source: Physical page 81, printed page 64; selected complete bibliographical note 3.6.

The notion of the adjoint subring of a fusion ring is motivated by the notion of the adjoint subcategory of a fusion category, introduced in [ENO2, 8.5], and developed in [GelaN]. The material on the adjoint subring, the universal grading, and nilpotent based rings is taken from [GelaN].

Source: Physical page 347, selected original bibliography entry [GelaN].

[GelaN] S. Gelaki and D. Nikshych. *Nilpotent fusion categories.* Advances in Mathematics **217** (2008), no. 3, 1053–1071.

Editorial note. The four book authors provide the collected exposition; the selected note preserves its explicit historical source credit. The complete printed proofs above keep their original compressed observations and final group-axiom check. The initial unital setup and subsequent general wording remain visible together, as do the 2015 dependency and the separate 2026 author clarification.